

The Fundamental Theorem of Calculus

MATH 797 — Week 1

Statement

Let f be continuous on $[a, b]$ and let $F(x) = \int_a^x f(t) dt$.

Then F is differentiable on (a, b) and:

$$F'(x) = f(x)$$

Furthermore, if G is any antiderivative of f :

$$\int_a^b f(x) dx = G(b) - G(a)$$

Key Insight

The theorem connects two seemingly separate operations:

- **Differentiation** — instantaneous rate of change
- **Integration** — accumulated area under a curve

They are inverse operations.